

Logic Bubble — Design Document (Structure & Skeleton)

1. Information Architecture

The application is structured around a public-facing marketing shell and an authenticated application core, with a single admin-only panel. All authenticated routes redirect unauthenticated users to the login page.

Site Map

```
Logic Bubble
■■■ Home (public)
■ ■■■ → Editor (CTA)
■ ■■■ → Dashboard (CTA)
■■■ Login (public)
■■■ Register (public)
■■■ Dashboard (authenticated)
■ ■■■ Create New Project
■ ■■■ Edit Project → Editor
■ ■■■ Delete / Export Project
■■■ Editor (authenticated, standalone page)
■ ■■■ Save / Export / New Project
■■■ Documentation (public)
■ ■■■ Sections: Getting Started, Features, Keyboard Shortcuts
■■■ About (public)
■ ■■■ Contact Form → /contact (POST)
■■■ Admin Messages (admin role only)
```

Navigation Zones

- Public nav (Home, Documentation, About, Login, Register): rendered in base.html, visible on all non-editor pages when logged out.
- Authenticated nav (Home, Documentation, About, username badge, Logout): rendered when a session is active.
- Editor topbar: a fully separate nav component; does not use base.html. Contains logo, mode indicator, project controls, and a gear preferences dropdown.
- Admin indicator: a mail icon appears in the nav when `session['role'] == 'admin'`, linking to the Admin Messages panel.

2. Database Schema

The application uses three tables in the logicbubbleapp MySQL database.

```
-- Registered users
CREATE TABLE users (
  id INT AUTO_INCREMENT PRIMARY KEY,
```

```

    username VARCHAR(80) NOT NULL UNIQUE,
    email VARCHAR(120) NOT NULL UNIQUE,
    password_hash VARCHAR(255) NOT NULL,
    role VARCHAR(20) NOT NULL DEFAULT 'user',
    preferences_json TEXT,
    created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP
);

-- User projects (canvas state + source text)
CREATE TABLE projects (
    id INT AUTO_INCREMENT PRIMARY KEY,
    user_id INT NOT NULL,
    title VARCHAR(200) NOT NULL,
    source_text LONGTEXT,
    graph_json LONGTEXT,
    created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
    updated_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE
CURRENT_TIMESTAMP,
    FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE
);

-- Contact form submissions
CREATE TABLE messages (
    id INT AUTO_INCREMENT PRIMARY KEY,
    name VARCHAR(100) NOT NULL,
    email VARCHAR(120) NOT NULL,
    subject VARCHAR(200),
    body TEXT NOT NULL,
    created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP
);

```

JSON Column Schemas

graph_json — Serialised canvas state stored as a JSON string:

```

{
  "bubbles": [
    {
      "id": "bubble-1",
      "x": 240,
      "y": 180,
      "text": "Node label",
      "color": "#7c6af7",
      "opacity": 1.0,
      "radius": 60
    }
  ],
  "dotNodes": [
    {
      "id": "dot-1",
      "x": 400,
      "y": 300
    }
  ],
  "lines": [
    {
      "id": "line-1",

```

```

    "sourceId": "bubble-1",
    "targetId": "dot-1",
    "style": "curved"
  }
]
}
```

preferences_json — Per-user editor preferences stored as a JSON string:

```
{
  "focusRingColorSelect": "#7c6af7",
  "focusRingColorConnect": "#c2e2ff",
  "fontFamily": "JetBrains Mono",
  "textFontSize": 14,
  "bubbleFontSize": 15,
  "lineStyle": "curved",
  "hintsVisible": true,
  "customShortcuts": {}
}
```

3. Page-by-Page Layout with Annotations

3.1 Home Page

Route: / — public, no authentication required.

Purpose: First impression and entry point. Converts visitors into registered users or returns authenticated users directly to the editor.

Layout:

[illegible]

Notes:

- "Open Editor" is the primary CTA; styled with --lb-accent background.
- "Go to Dashboard" is the secondary CTA; outlined style.
- If the user is already authenticated, both buttons route directly without an intermediate redirect.

- The mail icon in the nav is conditionally rendered server-side via Jinja2: `{% if session.get('role') == 'admin' %}`.

3.2 Dashboard

Route: /dashboard — authenticated.

Purpose: Project management hub. Users view, create, edit, export, and delete their saved projects.

Layout:

```

NAV: Logo | ... | username [logout icon]
H1: "Your Projects" [+ New Project] button
Project Project Project
Title Title Title
Created: Created: Created:
Updated: Updated: Updated:
[Edit] [Edit] [Edit]
[Export] [Export] [Export]
[Delete] [Delete] [Delete]
(empty state: illustration + "No projects yet")

```

Notes:

- Project cards displayed in a CSS Grid layout; responsive column count.
- "Edit" navigates to `/editor/<project_id>`.
- "Export" triggers a JSON download of the project's `graph_json`.
- "Delete" sends an AJAX DELETE request to `/api/projects/<id>`; on success, the card is removed from the DOM without a full page reload. A confirmation toast is displayed.
- Empty state is shown server-side when projects list is empty.

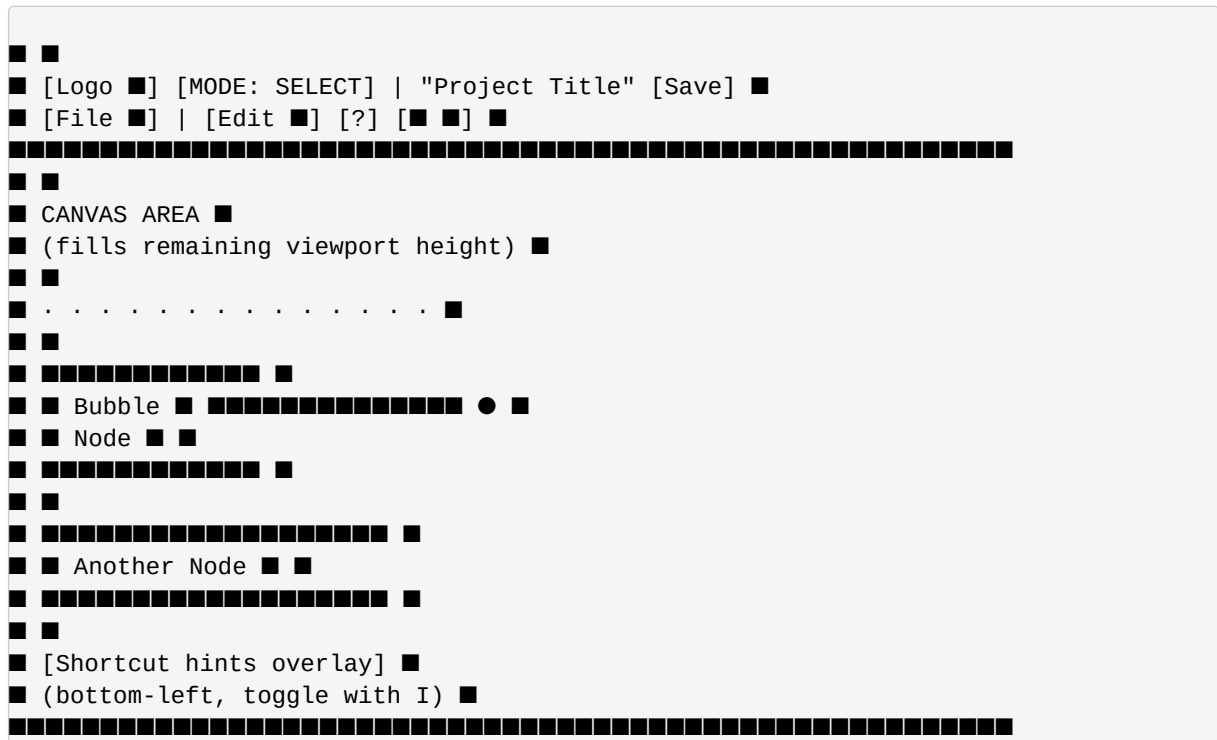
3.3 Editor

Route: /editor (new project) or /editor/<project_id> (existing) — authenticated. Standalone page; does not extend base.html.

Purpose: The core application. A full-viewport interactive canvas for building, connecting, and styling bubble-node diagrams.

Layout:

■ TOPBAR (fixed, full width, z-index high) ■



Topbar Dropdown Anatomy:

Dropdown	Contents
Logo	Links: Home, Dashboard, Documentation, About
File	New Project, Export as JSON
Edit	Undo (Ctrl+Z)
Gear	Toggle: Curve Lines; Toggle: Show Hints; — Font Settings (→ modal); Focus Ring Colour (→ modal); Keyboard Shortcuts (→ modal); — Logout

Mode Indicator Badge:

- Displays current interaction mode: SELECT (default) or CONNECT.
- Visual colour reflects the current focus ring colour preference.
- Toggled by pressing Tab or clicking the badge.

Canvas Interaction Zones:

- #canvas-container: the visible viewport; clips overflow; receives wheel and pointer events.
- #workspace: the pannable/zoomable child element; transform: translate() scale() applied via JS.
- Bubble nodes and dot nodes are absolutely positioned children of #workspace.
- SVG overlay: a full-size <svg> element covering #workspace containing all <path> connection lines and <defs> for arrowhead markers.
- Snap guide lines: temporary <div> elements appended to #workspace during drag, removed on drop.

Route: /documentation — public.

Purpose: Comprehensive feature reference for users learning the application.

Layout: Single-column scrollable page with a sticky or in-page nav for section anchors.

Sections:

- Getting Started — what Logic Bubble is, how to create a first project
- Canvas Navigation — pan, zoom, reset view
- Working with Nodes — bubble nodes, dot nodes, create/edit/delete/merge
- Working with Connections — creating arrows, toggling line style, retargeting, deleting
- Customisation — colours, fonts, focus ring, line style preferences
- Keyboard Shortcuts — full reference table (Key | Action | Mode)
- Saving and Projects — Ctrl+S, dashboard management, export

3.5 About Page

Route: /about — public.

Purpose: Project context and contact channel.

Layout:

[illegible]

Notes:

- Contact form POSTs to /contact.
- Flask handler validates fields and inserts a row into the messages table.
- On success: toast notification "Message sent." and form reset.
- On failure: inline error display.

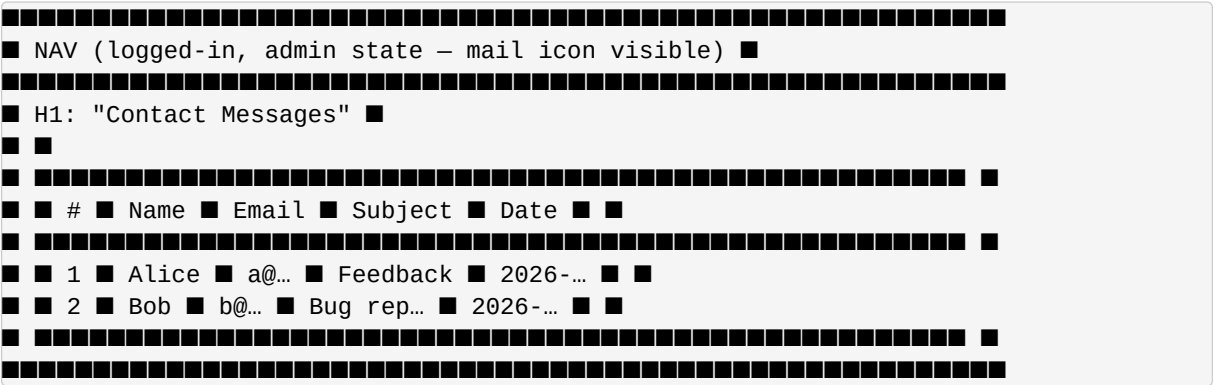
3.6 Admin Messages Panel

Route: /admin/messages — admin role only (session['role'] == 'admin').

Purpose: Review contact form submissions sent by users.

Access control: Flask route decorator checks role; non-admin requests receive a 403 response.

Layout:



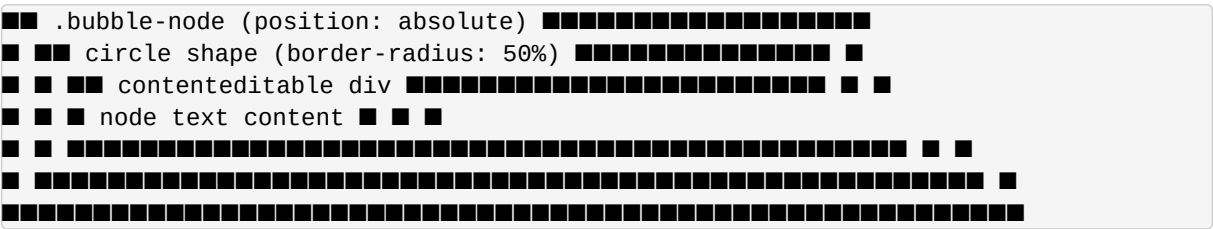
Notes:

- Table is rendered server-side via Jinja2 from a `SELECT * FROM messages ORDER BY created_at DESC` query.
- Message body is accessible via row expansion or a detail view (implementation-dependent).

4. Key UI Components

4.1 Bubble Node

Structure:



Sizing: Radius is computed from text length and font size to maintain a near-square text area. Minimum radius enforced to prevent degenerate shapes.

Visual States:

State	Visual Treatment
Default	Filled with bubbleColor, white/light text
Hover	Colour ring (border glow) appears around circumference
Focused — Select mode	Outer ring colour: focusRingColorSelect (default #7c6af7)

State	Visual Treatment
Focused — Connect mode	Outer ring colour: focusRingColorConnect (default #c2e2ff)
Dragging	Slight drop shadow; snap guides appear for alignment
Merge target (drag overlap)	Animated pulse to indicate merge will occur on drop

Editing: Double-clicking or pressing F2 with a node focused enables contenteditable. Pressing Enter or clicking away commits the edit and triggers a radius recalculation.

4.2 Dot Node

A minimal junction node for routing connection lines without label text.

- Shape: small filled circle (16–20px diameter)
- No text content
- Same interaction model as bubble node (drag, select, connect, delete)
- Created via right-click context menu → "Add Dot Node", or via a designated keyboard shortcut

4.3 Connection Line

Rendering: SVG <path> elements rendered inside the full-canvas SVG overlay.

Arrowhead: Defined as an SVG <marker> in <defs>. Marker ID referenced by marker-end attribute on each <path>.

Line Modes:

Mode	Path Type	Control
Straight	M x1,y1 L x2,y2	Default; or toggled via Shift+S
Curved	Cubic Bézier M ... C ...	Control points computed from node positions

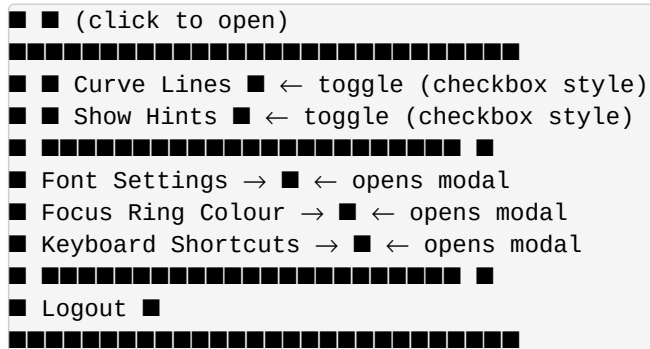
Interaction:

- Each rendered line has invisible hit-area padding for reliable click/touch targeting.
- Source handle: drag to a different node to retarget the connection's start.
- Target handle: drag to a different node to retarget the connection's end.
- Midpoint handle: drag away from any node to blank canvas space → connection is deleted.

4.4 Gear Dropdown (Preferences)

The gear (⚙️) dropdown in the editor topbar is the central access point for all editor preferences.

Structure:



Behaviour:

- Toggles apply immediately and persist to preferences_json via a debounced POST /api/preferences call.
- Clicking a sub-panel trigger opens the corresponding modal without closing or re-rendering the canvas.

4.5 Settings Modals (Font Settings / Focus Ring Colour)

All settings modals share a consistent three-zone layout:



Notes:

- Reset button in the header restores all controls in this modal to their default values (live, before Save is clicked).
- Cancel closes the modal and discards any unsaved changes.
- Save commits the changes to `preferences_json` and applies them to the canvas immediately.
- The live preview area reflects control changes in real time without requiring Save.

4.6 Right-Click Context Menu

Triggered by right-clicking on the canvas or a node. Content is context-aware:

On blank canvas:

- Add Bubble Node
- Add Dot Node
- Reset View

On a bubble node (select mode):

- Edit Text
- Change Colour (opens inline colour picker)
- Duplicate Node
- Delete Node

On a connection line:

- Delete Connection
- Toggle Line Style (Straight / Curved)

Implementation: A single absolutely-positioned `<div>` re-populated and repositioned on each right-click event; dismissed on any outside click, scroll, or Escape keypress.

4.7 Toast Notification System

Non-blocking status messages displayed in the bottom-right corner of the viewport.

- Stacked vertically (newest on top or bottom, configurable)
- Auto-dismiss after ~3 seconds
- Types: success (green accent), error (red accent), info (neutral)
- Announced to screen readers via an ARIA live region (`aria-live="polite"` for info/success, `aria-live="assertive"` for errors)

5. Colour & Typography

Design Token Reference

All colour values are defined as CSS custom properties in `theme.css` and referenced throughout `style.css` and component styles. Tailwind's config maps utility classes to these same `var()` references.

Token	Value	Usage
<code>--lb-bg</code>	<code>#0e1117</code>	Page and canvas background
<code>--lb-surface</code>	<code>#161b27</code>	Card backgrounds, modals, dropdowns, topbar
<code>--lb-surface-2</code>	<code>#1e2535</code>	Hover states on surface elements, nested panels
<code>--lb-border</code>	<code>#2d3650</code>	Dividers, input borders, card outlines
<code>--lb-accent</code>	<code>#7c6af7</code>	Primary interactive: buttons, active focus rings, links

Token	Value	Usage
--lb-accent-hover	#9585f9	Hover state for accent elements
--lb-text	#eef1f7	Primary body text
--lb-muted	#94a3b8	Secondary text, placeholder text, metadata
--lb-danger	#f87171	Delete actions, error states, danger toasts
--lb-success	#4ade80	Save confirmations, success toasts

Typography

Context	Font Family	Weight	Notes
Site headings	Syne	700	All page H1–H3, hero title
Site body / nav	Syne	400	Navigation links, paragraph text
Editor topbar	JetBrains Mono	400–500	Topbar labels, mode badge, project title
Bubble node text	User-selected (default: system-ui)	400	Configurable via Font Settings modal
Code blocks (docs)	JetBrains Mono	400	Documentation page only

Font sizes are defined as --lb-text-sm, --lb-text-base, --lb-text-lg, etc. tokens in theme.css, ensuring consistent scaling across components.

6. Accessibility Notes

Accessibility is treated as a baseline requirement, not an enhancement. The following practices are applied throughout:

Area	Implementation
Keyboard navigation	All interactive elements (buttons, links, inputs, canvas nodes) are reachable and operable by keyboard. Tab order follows logical reading order.
Focus visibility	Custom focus ring styles are applied globally via --lb-accent; editor nodes use the user-configurable focusRingColor tokens. Focus outlines are never suppressed without a visible replacement.
Colour contrast	Primary text (--lb-text on --lb-bg) targets WCAG AA (4.5:1). Accent interactive elements are tested against their backgrounds.
Semantic HTML	<nav>, <main>, <header>, <footer>, <button>, <label>, <form> elements used throughout. Div soup avoided in structural markup.

Area	Implementation
ARIA live regions	The toast container is marked <code>aria-live="polite"</code> to announce non-critical notifications; error toasts use <code>aria-live="assertive"</code> .
Form labels	All form inputs have associated <code><label></code> elements (either explicit <code>for/id</code> pairing or wrapping label).
Modal focus trap	When a modal is open, keyboard focus is trapped within it. Escape key closes the modal and returns focus to the triggering element.
Alt text	Decorative images use <code>alt=""</code> . Meaningful images carry descriptive alt text.

7. Responsive Scope

Logic Bubble v4.1.x is a desktop-first, desktop-only application.

Rationale: The core canvas interaction model — pan with mouse drag, zoom with scroll wheel, precise node positioning, drag-to-connect — is fundamentally dependent on a mouse pointing device. Touch and stylus equivalents would require a separate interaction design and are deferred to a future version.

Current responsive behaviour:

Viewport Range	Behaviour
≥ 1024px (desktop)	Fully supported; all features functional
768px – 1023px (tablet)	Navigation collapses gracefully; public pages (Home, About, Docs) remain readable; Editor may be constrained but is not blocked
< 768px (mobile)	Public pages readable; Editor not recommended and not tested on touch devices

Implementation: CSS breakpoints handle nav collapse and card grid reflowing on narrower viewports. The editor canvas does not apply mobile-specific overrides; it degrades gracefully but is not optimised.